Enhancing Green Roof Design in Modern Architecture through Advanced BIM Integration

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I. Abstract

This dissertation focuses on applying BIM in the creation of green roofs in contemporary architectural design to improve such aspects as environmental sustainability, economic performance, and social impacts. Literature survey reveals the trends, gaps, and findings in the research literature concerning the applications of BIM for green roofs. Real life examples of BIM application in different projects such as Bloomberg London, The Edge in Amsterdam, Lagg Distillery in Scotland, and Forester hill Health Campus in Aberdeen is described in detail in this book. Based on this analysis, it is possible to state that BIM has shown to be an effective technique for support designers in enhancing design accuracy, increasing the overall project effectiveness and promoting the sustainable development of green roofs. These findings further strengthen the framework intended for architects, engineers, and urban planners to enhance the application of BIM for green roof projects and, thus, promote sustainable urban system.

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III. Declaration

This dissertation is the outcome of the author's own research. It was written by the author and was not previously submitted for examination, resulting in the award of a degree. The author owns the copyright to this dissertation under the rules of the United Kingdom Copyright Acts, as specified in University of Strathclyde Regulation 3.50. Any use of material included in or derived from this Dissertation must be acknowledged.

IV. Abbreviations

AI - Artificial Intelligence

BIM - Building Information Modelling

BREEAM - Building Research Establishment Environmental Assessment Method

COBie - Construction Operations Building Information Exchange

CFD - Computational Fluid Dynamics

DBB - Design-Bid-Build

EIR - Employer's Information Requirements

FM - Facility Management

GIS - Geographic Information Systems

HVAC - Heating, Ventilation, and Air Conditioning

IoT - Internet of Things

ISO - International Organization for Standardization

LCA - Life Cycle Assessment

LEED - Leadership in Energy and Environmental Design

MEP - Mechanical, Electrical, and Plumbing

O&M - Operations and Maintenance

PIM - Project Information Model

PM - Project Management

RFI - Request for Information

ROI - Return on Investment

SLA - Service Level Agreement

SBS - Sick Building Syndrome

SDGs - Sustainable Development Goals

UAV - Unmanned Aerial Vehicle

UK - United Kingdom

USGBC - U.S. Green Building Council

VDC - Virtual Design and Construction

VR - Virtual Reality

WBS - Work Breakdown Structure

Chapter 1: Introduction

Currently, green roofs are seen to be an indispensable component of modern urban environment with possible benefits of ecological, economical, and social nature. Such systems are advantageous for urban areas as it reduces the rate of stormwater runoff, cools urban is-lands, and enhances air quality. Oberndorfer et al. (2007), (Tighnavard Balasbaneh et al., 2024). BIM instrumentation aids in designing environmental and structural green roof structures by developing a better simulated environment(Allouhi et al., 2023). Nonetheless, it has been stated that BIM has not been effectively applied in integration of green roofs because there are no special procedures set for green roof, especially for addressing their special requirements that include variety of plant species and complex watering systems. However, the implementation of a BIM system in green roof projects has its significant barriers as follows; Specifically lacking are generic approaches that would interlink the application of BIM with the attributes of green roofs in cities(Williams et al., 2010). The goal of this research is to address the following questions regarding BIM in green roof design for enhancing environmental, economic, and social factors in urban architecture: The first chapter of this work shall be the introduction and shall include the following subtopics: Overview of the study, statement of the research problem, the research objectives, the hypothesis of the study, study goals, the value of the study, study limitations, and a summary of the research.

1.1 Background of the Study

Energy efficient systems integrated into various cities defined as green roofs, are considered significant for the enhancement of the sustainable construction processes and embrace various values such as the storm water retention the temperatures moderation, and improved air quality(Oberndorfer et al., 2007, Smith and Sajjadian, 2024). Green roof environment and structure can be more effectively maintained by enhancing the BIM application to analyse the objects' design(Getter and Rowe, 2006). However, BIM is not yet maximally utilizing green roof integration because there is no specific protocol for using BIM with a system as multifaceted as green roof integration with multiple plant species and a highly systems using water(Smith and Sajjadian, 2024, Williams et al., 2010). These are as early as the hanging gardens of Babylon though further development was done in the 1960s and 1970s in Germany(Francis and Jensen, 2017). From the recent literatures one understands that cities like Chica-go, Toronto and Singapore are promoting Green roofs to reduce the environmental concerns of urban society (Raimondi and Becciu, 2021).

1.2 Statement of the Research Problem

Presumably, BIM can be integrated with green roof projects, however, there is a poor understanding, if current strategies of BIM are efficient in handling the environmental issue of minimizing urban landscapes and the operational issues related to green infrastructure. Even when it comes has more specificity to various projects within the city's architectural context, there is

variation in how BIM has been applied and, thus, it is harder to distil best practices or properly assess the actual extent of improvements in sustainability and efficiency.

1.3 Research Aims, Objectives and Questions

I. Aim

The aim of this study is to thoroughly investigate possibilities for the most effective approach to integrating BIM for green roof design in urban architecture to assist the highest possible environmental, economic, and social value, and thereby optimizing the sustainability and functionality of urban buildings.

II. Objectives:

1. 1. To examine the use of Building Information Modelling in designing and developing green roofs for diverse architectural projects.
2. To identify the environmental, economic, and social benefits realized using BIM in green roof designs.
3. To evaluate BIM's influence on building lifecycle management for green roofs.

III. Research Questions

1. *How is BIM currently integrated in the design and development of green roofs across various architectural projects?*
2. *What The socioeconomic, environmental, and cultural benefits realized through the integration of BIM in green roof designs?*
3. *What impact does BIM have on the lifecycle management and operational efficiency of buildings with green roofs?*

1.4 Significance of the Research

This study seeks to improve the integration of Building Information Modelling with green roof projects in urban architecture. By refining BIM strategies, it intends to promote the widespread implementation of Green Roofs, enhance urban sustainability, and improve building performance. The study will develop standardized guidelines for incorporating BIM in green roof designs, shaping policymaking, establishing sustainable construction best practices, and driving urban sustainability progress.

1.5 About the Dissertation

This research work aims at exploring the effectiveness of BIM in improving the design, construction, and lifespan of green roofing in the architectural construction projects in urban areas. This paper also elaborates on how BIM can be better incorporated for achieving the holistic objectives on Green roofs' environmental, economic, and social implications. Thus, the research aims at identifying procedures that can enhance the implementation of green roofs by examining the current applications and drawbacks in BIM strategies. The dissertation has formulated several research questions through which it aims at elaborating the integration of BIM

in green roof designs, the benefits accruing there from, and the implications on building lifecycle management. The conclusions are intended to improve BIM implementations in green roof projects, raise the standards of environmental management, and optimize the built environment results of various cities. Thus, based on a systematic literature review with multiple case studies, this study offers valuable findings on the possibilities of BIM to improve urban environments sustainably.

Chapter 2: Literature Review

2.1 Green Roofs, Their Types, and Current Trends

Green roofs also called vegetative roofs or eco-roofs are roofs which are partially or fully covered with plants growing over a water proofing layer(Shafique et al., 2018, Korol and Shushunova, 2016). They often include other layers such as root barrier, drain layer, and irrigation layer. Green roofs can be as simple as applying sedum in the structure, or as complicated as building actual gardens within the roofs with different types of plants present(Teotónio et al., 2021). They are proposed as sustainable in terms of environmental, economic and social returns as converts of otherwise vacant roof areas into live green roofs.

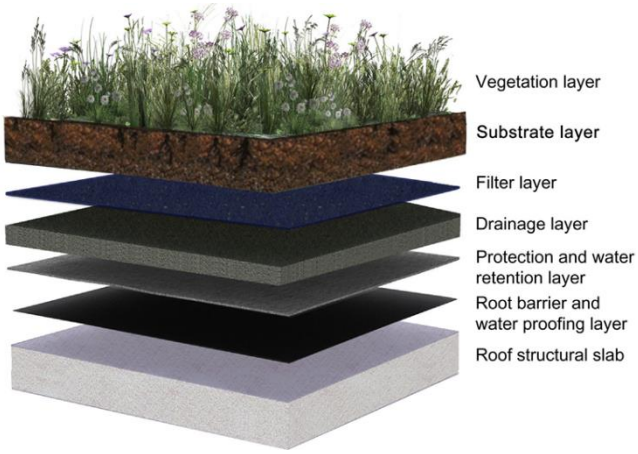


Figure 1.Green Roof Layers

2.1.1 Why are Green Roofs Used

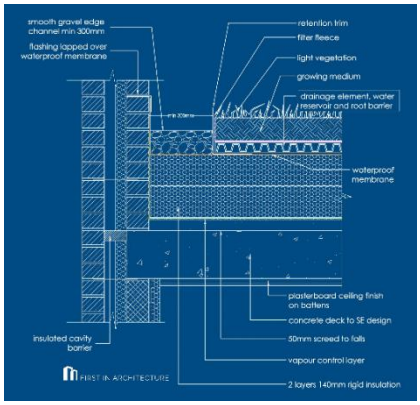


Figure 2. Green roof Section

Green roofs are used for several reasons, primarily for their Sustainable advantages. They assist reduce the urban heat island effect. by cooling the surrounding air and providing natural insulation, which reduces the energy needed for heating and cooling buildings (Raimondi and Becciu, 2021)Additionally, green roofs improve air quality by filtering pollutants and capturing airborne particles. They also serve a vital role in stormwater management by absorbing rainfall, reducing runoff, and decreasing the demand on urban drainage networks (Getter and Rowe, 2006, Sysoeva and Bogachev, 2023).

In the economic aspect, green roofs help in conserving the roofing materials by shielding them from ultra violet radiation and high heat variation; this consequently decreases costs on maintenance and replacement. (Bianchini and Hewage, 2012). In social context, green roofs beautify

the urban environment, offer recreational places for people, and positively affect the inhabitants' quality of life as they offer green urban territories for relaxation and communication.(Oberndorfer et al., 2007, Kotzen, 2018).

2.1.2 Where are Green Roofs Used?

Green roofs are being implemented in a variety of settings, from residential and commercial buildings to public institutions and industrial facilities(Semaan and Pearce, 2016, Xiao et al., 2014). They are particularly beneficial in densely populated urban areas where green space is limited. Cities all over the world, including Chicago, Toronto, and Singapore, have adopted green roof initiatives to promote sustainable urban development(Dvorak, 2009, MacIvor et al., 2016),(Low, 2011) These cities often provide incentives, such as tax benefits or subsidies, to encourage building owners to install green roofs (Gargari et al., 2016).

In addition to urban environments, green roofs are used in educational institutions, healthcare facilities, and corporate campuses(Kovacs, 2017), where they can serve educational purposes(Díaz-López et al., 2022), enhance the healing environment, and improve employee productivity and satisfaction. Green roofs are also gaining popularity in industrial settings, where they help manage stormwater runoff and reduce the heat load on buildings, contributing to overall operational efficiency(Scolaro and Ghisi, 2022).

2.1.3 Types of Green Roofs

Green roofs are becoming an integral part of sustainable urban architecture, providing a multitude of benefits that address various environmental, economic, and social challenges in cities(ziaee et al., 2022).

This section discusses the three primary categories of green roof systems

—**extensive, intensive, and semi-intensive**—highlighting their characteristics and current trends(Yildirim et al., 2023, Manso et al., 2021).

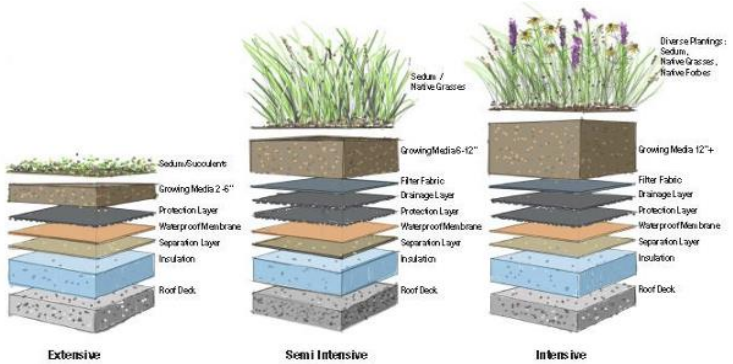


Figure 3.Types of Green Roofs

1. **Extensive Green Roofs:** These are distinguished by low substrate depth (generally 100-150mm) and are of low weight (70-170 kg/m²)(Yildirim et al., 2023). Green roofs are generally low in their maintenance needs as the extensive ones are densely covered with sedum,

mosses, and grasses and very limited plant varieties. They are mainly not intended for leisure trips and are suitable for constructions that can hardly support large weights because of their low densities. These roofs offer the following ecological gains; they Minimize the urban heat island effect, enhance air quality, and also act as insulators(Yildirim et al., 2023).

2. Intensive Green Roofs: Thus, intense green roofs are defined by a substrate layer of at least 200mm, with the weight of up to 290-970 kg/m², which allows for planting tree, shrubs and flowers(Perez and Coma, 2018). These roofs working at their full potential demand considerable maintenance, owing to the variation in plethora of plant types; these rooftops are mostly intended as accessible ones, for cases such as the rooftop farming [28, 29]. Apart from being cost effective solutions to ecological problems they also offer social and recreational places for habitats of the urban areas beautifying the look and feel of urban space as well as increasing the bio-diversity(Yildirim et al., 2023).

3. Semi-Intensive Green Roofs: Produce these roofs has an intermediate depth of substrate (120-250mm) they intermediate between the characteristics of large and dense green roofs. They have a moderate weight and the level of attention they require being moderate, they allow a range of plants, including grasses, herbs, as well as small shrubs(Pérez and Coma, 2018). Semi intensive green roofs with partial access are partially open to human activities but could support the ecosystem and have positive impacts on the society hence can be applied in most cities(Yildirim et al., 2023, Pons Fiorentin et al., 2024).

Feature	Extensive green roofs	Intensive green roofs	Semi-intensive green roofs
Substrate Depth	100-150mm (shallow)	200mm or more (deep)	120-250mm (intermediate)
Weight	70-170 kg/m² (lightweight)	290-970 kg/m² (heavier)	150-400 kg/m² (moderate)
Maintenance	Low maintenance	High maintenance	Moderate maintenance
Plant Diversity	Limited (sedum, mosses, grasses)	High (trees, shrubs, flowers)	Intermediate (grasses, herbs, small shrubs)
Accessibility	Usually not accessible	Often accessible as roof gardens	Partially accessible
Ecological Benefits	Good for insulation, reduces urban heat island effect, and improves air quality	Enhances urban biodiversity, better stormwater management, and air quality improvement	Provides moderate ecological benefits, balancing extensive and intensive features
Economic Benefits	Lower installation costs, long-term energy savings, extended roof lifespan	Higher installation costs but can increase property values and provide significant energy savings	Moderate costs and benefits, suitable for buildings requiring intermediate solutions
Social Benefits	Minimal due to limited accessibility	High due to accessibility, recreational space, and community engagement	Moderate, providing some communal space and aesthetic benefits
Typical Use Cases	Buildings with limited load-bearing capacity, focus on ecological benefits	High-value buildings, urban areas needing recreational spaces, and enhanced biodiversity	Mixed-use buildings requiring both ecological and social benefits

Figure 4.Comparative analysis of Types of Green Roofs

2.1.4 Current Trends in Green Roofs

Various studies show that green roofs have been embraced because of the enhanced focus on the aspect of sustainability in the cities. Recent trends include:

1. **Integration with Urban Infrastructure:** They have become common features of modern design among the public facilities and commercial structures to reduce the Urban Heat Islands and for storm water control(Shafique et al., 2018).

2. **Technological Advances:** Innovations in green roof technology for instance; automated irrigation systems and light weight substrate to help facilitate green roofs for optimal performance(Scolaro and Ghisi, 2022).
3. **Policy and Incentives:** The current policies in many cities have called for compliance and offering of incentives for construction of green roofs. This comprises of cash incentives, policy exemptions, and legal requirements(Bernegger et al., 2021).

Thus, the green roofs do not only bring numerous ecological values but economic and social values as well, which makes them a crucial aspect in the long-term development of urban territory.

2.2 The Impact of Green Roofs in Urban Architecture

Figure 5.Key aspects of Green Roofs

Hence, green roofs are a vital part of the current green and sustainable urban architectural structure since they spur environmental improvements, financial returns, and social benefits. Some of the benefits that relate to urban environments include maintaining and reducing heat islands in urban built environments, controlling air quality, and as well, managing storm water(Pons Fiorentin et al., 2024, Silva et al., 2023). However, green roofs are not without some problems which are associated with introduction of these modern facilities in urban environment. For instance, the costs of installing green roofs can be very high, and secondly the green roofs require complicated process of maintenance. Through integration of BIM in the planning, installation and maintenance of green roofs, they become more feasible and productive(Marín et al., 2023).

2.2.1 Ecological Benefits

The adoption of green roofs can be attributed to the following ecological benefits. They help to mitigate the most severe impacts of the growing urban heat island effect by providing insulation and through evapotranspiration. This process significantly reduces the temperature of the air around the roofing structure reducing energy usage by nearby structures(Czemiel Berndtsson, 2010). In the same way, green roofs also contribute to the increase of species diversity within urban for goodness of both plant and animal species hence helping improve ecosystem in such places. besides, clean the air, thus helping to eliminate contaminants such as particles and carbon dioxide, making cities more suitable for people to live in(Lee et al., 2024).

2.2.2 Economic Benefits

Green roofs have enormous economic benefits, particularly in terms of long-term expenses. Although the upfront investment for green roofs is quite expensive, they help to conserve energy through improved insulation thus reducing on the costs of heating and cooling. (Teotónio et al., 2018, Feng and Hewage, 2018). Studies demonstrate that buildings with green roofs have a longer lifespan due to the protection they offer to the roofing materials, reducing maintenance and replacement costs over time (Bianchini and Hewage, 2012, Feng and Hewage, 2018, Lee

et al., 2024). Moreover, green roofs can increase property values and attract more tenants or buyers interested in sustainable living spaces (Scolaro and Ghisi, 2022, Shafique et al., 2020).

2.2.3 Social Benefits

From a social perspective, green roof benefits the The lives of individuals living in urban areas through provision of the green areas for leisure and recreation. Such areas make for enhanced mental wellbeing of people within cities for it provides a relief from the concrete jungle(Kotzen, 2018). Moreover, as found by Oberndorfer (Oberndorfer et al., 2007), green roofs cut down on the overall social exclusion owing to provision of community gardening or other related uses. They also offer education concerning practice and implementation of urban agriculture and sustainable use of resources (Alasmari et al., 2023),(Antón and Díaz, 2014).

2.3 Introduction to Building Information Modelling (BIM)

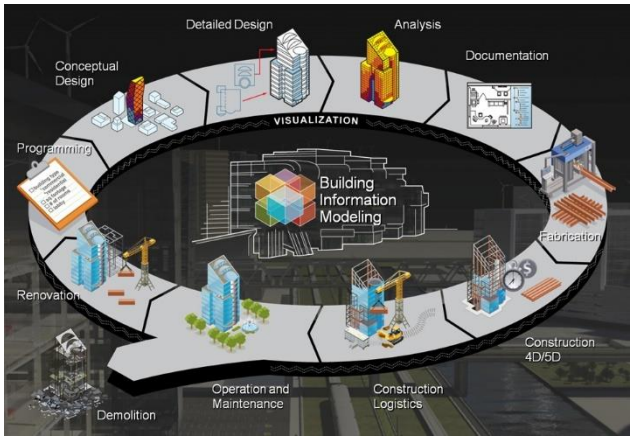


Figure 6. Building Information Modeling

BIM is described as the process of employing a digital source of the physical and functional features of a facility for the development, management, and exchange of information that is important for every phase of a facility, from conception to deconstruction. BIM enhances the production of intelligent 3D models that incorporate data from various fields such as architectural, engineering and constructional, making available up-to-date information to all the stakeholders, thereby enhancing teamwork and minimizing on mistakes (Waqar et al., 2023, Antón and Díaz, 2014). Most importantly, it provides lifecycle management as detailed database keep records of how facilities are being managed and possibly used in future renovations. Organization benefits from BIM in a way that Model simulations and analyses contribute to increased design quality and evaluation of the energy performance, light, sound, and structural preciseness(Onososen et al., 2022, Lu et al., 2017). The cooperative environment thus eliminates unnecessary expenses and time use, and guides architecture towards eco-friendly sustainable design through enhancement of the building's energy use. Moreover, BIM also helps to eliminate the need for drafting and creation of construction documents, which otherwise can create inconsistencies in the content; in addition, BIM improve the standards of project quality and communication among the related stakeholders(Azhar, 2011, Casini, 2022),(Datta et al., 2023).

2.4 BIM and Architectural Design

BIM in the United Kingdom has brought changes to the architectural design and project management of facilities where one can get the following benefits for the improvements of the various aspects of constructional process.

This section will discuss five key advantages of BIM: This includes *better visualization, more accuracy, detection of clashes, better decisions making, and pinpoint Project management*(Nguyen et al., 2010). BIM enhances visualization since it makes it possible for a client or any stakeholder to get better visualization of what the building will look like since 3D models can be easily developed. It improves accuracy because changes to the model are communicated to all related sub-systems, thus minimizing mistakes(Chen, 2023, Alasmari et al., 2023). The technology integrated into the system also could detect clashes that are likely to occur throughout the building process and the designer is normally advised to make changes in the design to avoid such incidences(Alasmari et al., 2023).

In addition, it connects different type of data according to, for example, cost and energy performance data, thus assisting in enhancing design quality and project outcomes. Furthermore, that BIM allows project development to produce construction documents and schedules for efficient project coordination and timely completion(Volk et al., 2014, Zimmermann et al., 2021). These benefits muddled show that BIM is revolutionizing the existing AEC environments within the current architecture practice(Azhar, 2011).

2.4.1 Enhanced Visualization

Overall, it can be said that BIM considerably enriches architectural designing in the UK with regards to the guidance of visualization. BIM technology permits architects to produce highly detailed and accurate 3D models that go as far as to include geometries of building, details of the materials to be used, the system to be incorporated as well as the components. The kind of visualization leads to improved communication since clients and other stakeholders can be in a better position to understand the proposed designs(Alasmari et al., 2023, McArthur, 2015).

For instance, in ***the Crossrail project***, BIM was applied to create detailed information models which enabled the integration of required complicated str

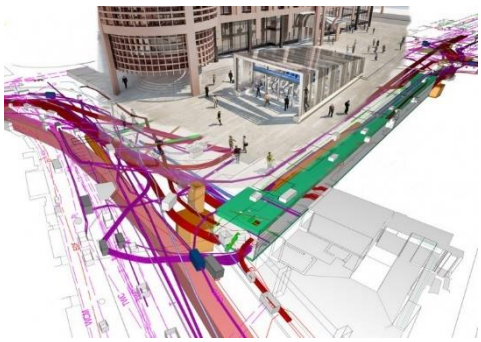


Figure 7.Cross Rail Project

2.4.3 Clash Detection

BIM is another technique that has a strong feature of clash detection that helps in identifying the interference between the systems in the planning stage rather than the construction stage.

This feature proved to be especially useful in the ***Queen Elizabeth Olympic Park*** project for the London 2012 Olympics. The use of BIM application helped in the detection

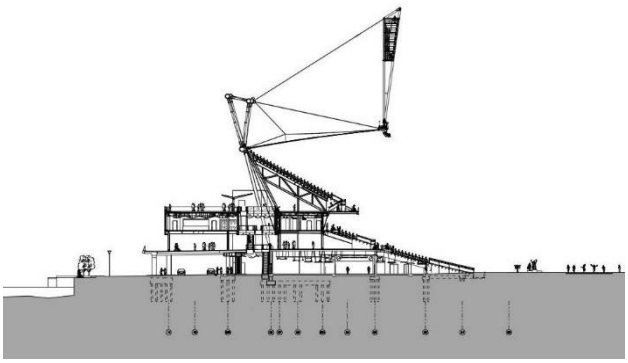


Figure 8.Part Section of London Queen Elizabeth Olympic Park

and solving of interferences between the mechanical, electrical, and plumbing systems at a tender stage. Thus, elaborating these problems in the design stage helped the project to minimize construction time and costs, thus improving the overall construction process. Thus, the use of BIM and its clash detection capabilities helped avoid costly changes and delays During the development phase of the construction, large and complex projects in general(alliesandmorrison).

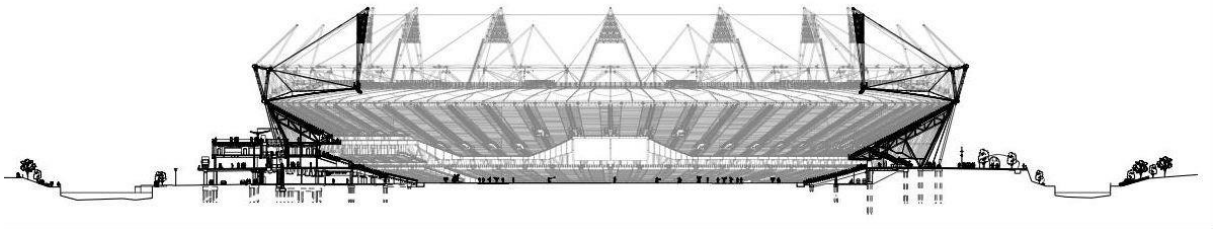


Figure 9. Section of London Queen Elizabeth Olympic Park

2.4.4 Informed Decision-Making

The use of BIM in incorporating different data such as cost and energy data into the design process has a positive impact on the design and performance of projects. A great example might be found in the ***King's Cross development in London***. The architectural design of the buildings during redevelopment was done with the use of BIM where measures that improved the energy performance of the buildings were included. Thus, by applying BIM to the energy efficiency assessment, the project succeeded in enhancing the sustainability of the design and proving how BIM can help in the development of environmentally responsible urban environments. The incorporation of such data within BIM meant that right decisions were made in relation to the different aspects of the project, hence resulting in better project performance, thus pointing to the value of BIM in current architectural practices(alliesandmorrison).

2.4.5 Efficient Project Management

The implementation of BIM results in effective organization and control of construction projects since it provides accurate construction documents and schedules. This capability helps to achieve the intended objectives of projects and to execute each element of the project to the letter. An example of this is the *West Cambridge Development by the University of Cambridge*. This project also applied BIM in generating the right schedule and coordinating the special aspects of the project(Wang et al.). This cemented that the project was executed on the schedule and below budget, demonstrating that BIM is a helpful tool in controlling the development process particularly huge projects. In this way, due to implementation of the BIM, the West Cambridge Development project reached

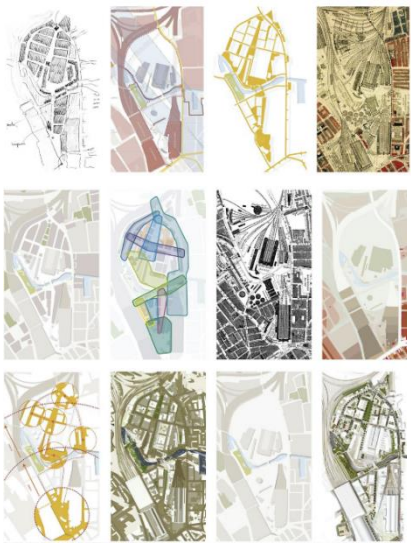


Figure 10. King's Cross Development Using BIM

its objectives and targets in a most efficient manner(Marcellino et al., 2023).

2.5 Previous Integrations of BIM in Green Roof Projects

Several examples have been presented in the literature on how BIM has been used effectively in green roof projects: therefore, the effectiveness of building projects.

Figure 11.33 Central in the City of London

For instance, the ***33 Central in the City of London*** that features one of the largest green roofs in the city credited BREEAM Excellent to the use of BIM in the management of strategies and designs of phases and the HVAC position that maximized the green roof capacity.

As with the ***Maggie's Centre*** case study, this Wolverhampton project also incorporated the green roof design concept whereby BIM was employed to assess the environmental performance of designs and to confirm adequate system integration.

The Crystal in London, one of the sustainable cities' projects by Siemens, incorporated the utilization of the Building Information Modeling software may improve energy consumption and also in the way storm water is dealt with through the integration of the green roof to the other facilities in the engineering process. The above examples demonstrate how application of BIM can greatly influence the planning and execution of green roof projects and could assist to improve the outcomes of the project and altering the ways of handling the sustainability of buildings.

Figure 12.Crystal, London

2.6 Gaps in Current Research and Practice

2.6.1 Limited Integration of BIM in Small-Scale Projects

The use of BIM is, however, less common in the cases of small-scale projects despite its frequent application in the large and famous projects. The possible reasons for this gap may include the following: BIM has been viewed to be costly and complicated to implement. That means small architectural firms cannot fully leverage on BIM due to the resource and skill

limitations, hence, cannot generate multiple benefits across multiple projects. For instance, the Crossrail project in London which is a large-scale project used BIM to increase the control of the complex structural graphics and the early design changes while small-scaled project lacks the resources to purchase BIM software and train its employees. For instance, consider a relatively basic residential refurbishment project where BIM may get in the way of sequencing and improve results but is not used due to these issues. The possible BIM application concerning a small-scale project with a simplified process flow chart will describe the benefits of the application and the need for further development in detail(Akbari et al., 2024).

2.6.2 Limited Focus on Lifecycle Management

Current BIM practices are mainly directed towards the design and construction of the facilities, particularly, without regard to the operational and maintenance stage of a facility. Even though The usage of BIM has increased dramatically in the building sector, there is a poor literature on how the same can be applied effectively in facility management and in providing routine maintenance. BIM along with other smart technologies like the IoT, if implemented, can be more useful in the management of building operations but this area is still relatively unexplored.

For example, ***Bloomberg London HQ*** is a new construction project that adopted BIM in its design and construction but its application in maintenance is scarce. This gap implies that BIM may be useful during many phases of a building's life, such as energy transfer and repair. A good example can be explained with the use of BIM that manages the structure of Sydney Opera House using IoT sensors. Maybe, it is possible to supplement the description of these extended benefits by a simple flowchart that would show the integration of BIM with the lifecycle management processes(Zeiss, January 02, 2019).

Figure 13. Bloomberg, London HQ

2.6.4 Research Gap Summary

The research and practice gaps that are noted in this study therefore determine the areas that need revision in regard to the integration of BIM. To address the challenge of BIM's limited use in small construction projects, enhance better standardization and integration, and expand the focus on the lifecycle is crucial in getting more value out of BIM. The following will increase the benefits of BIM in the delivery of higher efficiency, sustainability and cost effectiveness in building and construction projects of all scales, types, and uses.

Chapter 3: Methodology

This chapter outlines the research strategy and methodology utilized to meet the study's objectives. The objective of this research is therefore to critically assess and improve the implementation of BIM in green roof systems in urban architecture and thus contribute to the environmental, economic and social performance of buildings.

The research questions are: How is BIM applied in the design and construction of green roofs? What are the outcomes of adopting BIM in the case of green roofs? To what extent has the involvement of BIM affected the management of buildings with green roofs?

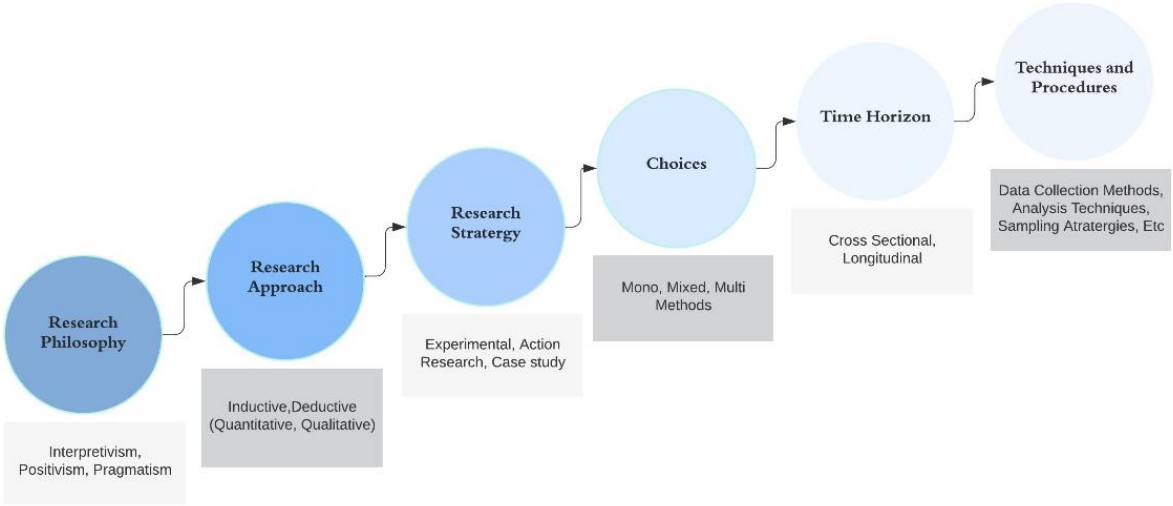


Figure 14..Saunders' Research Onion Diagram (Author's Own, 2024)

To attain these goals, this chapter discusses the chosen study's philosophy, approach, strategy, and time horizon as postulated by Saunders' Research Onion diagram(Seuring et al., 2021). Thus, the research philosophy chosen is Interpretivism. Hypotheses on BIM integration in green roofs have been tested using deductive strategies used in this study. The research design applied in this study is case study, which aims at exploring the practical use of BIM in detail. Based on the cross-sectional design, a time horizon of the present is selected to collect data on practices and results at a particular moment. Thus, with the help of these methodologies, the research question on improving BIM integration in green roof projects for urban sustainability is explored in a systematic manner(Fellows, 2015).

3.1 Research Philosophy

Interpretivism is a research philosophy that involves the study of people's behaviour, and social phenomena based on the interpretation of the meanings they attach to their actions in their own social context. This approach allows for the uniqueness of the individual and his perception of the world around him. Interpretivists tend to utilize the qualitative data collection and analysis

techniques like interviews, observations, and text analysis to understand the real-life phenomena in their entirety(Fellows, 2015).

Regarding the research topic *"Enhancing Green Roof Design in Modern Architecture through Advanced BIM Integration,"* an interpretivist philosophy is most relevant. This research work will seek to evaluate and enhance the implementation of BIM in green roof design within the urban architectural structures with the view of increasing the environmental, economic and social returns for improved sustainability and performance.

3.1.1 Why Interpretivism?

- **Understanding Contextual Nuances**: The projects can be quite diverse based on the location within the city, climate, and the characteristics of the building. An interpretivist approach enables the researcher to examine such differences and variations in the use of BIM in different architectural projects.
- **Qualitative Methods**: The research objectives and the questions proposed are most suitable for qualitative examinations. For example, the research objectives that can be investigated with the help of detailed interviews and case studies include the integration of BIM (Objective 1) and the assessment of the lifecycle management impact (Objective 3). These methods will reveal some patterns that the quantifiable data may not show.

3.1.2 Applications of Interpretivism in the Research

- **Data Collection Methods**: The research will be based on case studies and comparisons of several green roof projects that have used BIM. This technique allows for the detection of the degree to which BIM is used in diverse contexts, as well as the subsequent implications. Thus, through the analysis of several cases, the research will demonstrate the specifics of the integration process and explain how BIM can be applied in various contexts.
- **Outcome**: As a result of choosing the interpretivist research philosophy, the research seeks to come up with detailed descriptions of the integration of BIM in green roof projects. These will help inform the creation of standard procedures for integrating BIM into green roof designs to achieve the best environ-mental, economic, and social results. The comparison of the case studies will show the successes and pitfalls of the measures, which will be useful for future projects of this kind.

3.2 Research Approach

3.2.1 Why Inductive Approach?

This research approach is most suitable in situations where the research questions are geared towards understanding and establishing the behaviours of events and come up with new knowledge. The goal of this research is to analyze the deployment of BIM in green roof projects, as well as the potential and problems that come from it. This means that a more relaxed approach is needed that would enable identification of new trends. Some of the green roof projects' components vary depending on the climate, city planning, and some features of the

construction of the building. Thus, using an inductive approach, the researcher can probe these specific circumstances and get a better understanding of how BIM integration applies in specific cases. Deductive approach is conducted in a way that the researcher has a given hypothesis which he or she wants to test while in the inductive approach, theories are developed from the data collected(Fellows, 2015).

This is crucial for this study because it aims at developing standard guidelines on how to incorporate BIM in green roof projects and the findings of several case studies will be used in this regard. Hence, it is feasible to conduct the research on real-life projects and derive recommendations for the successful implementation of sustainable as well as operationally reliable urban buildings.

3.3 Research Strategy

Using case studies is more appropriate in this research since it reveals detailed information on how BIM is applied in green roof construction. Thus, the selection of the study methodology and plan gives a thorough understanding of the depth and context of integrating BIM with green roofs. This approach enables the researcher to get a clearer picture of the different experiences that are present in green roof projects among the different cities, the challenges that are encountered, and the outcomes of the BIM integration in green roof projects.

Since the research objectives of the paper focus on the analysis of the BIM integration to enhance its environmental, economic, and social impacts, case studies will help identify the best practices and trends. Thus, based on the analysis of several cases, the research can identify best practices that would be relevant and effective in increasing the sustainability and operational efficiency of urban buildings. This method corresponds to the proposed research objectives as it enables the generation of extensive and detailed data that reflects the complexity of the specific context and can be used to provide effective recommendations regarding the integration of BIM into green roof design(Fellows, 2015).

3.4 Data Collection and Analysis

As part of my dissertation, I conducted a comprehensive literature review to explore the application of Building Information Modeling in green roof design in today's architecture. A systematic literature review is a step-by-step procedure of identifying, critically evaluating, and integrating previous work to answer a specific research question. This helped me to understand research trends, areas of research that needs to be filled and results of the prior researches on BIM and green roofs(Fellows, 2015).

3.4.1 Research Questions and Objectives

Thus, to structure the review, I formulated the following research questions and objectives that would help to identify how BIM is applied in green roof projects, the advantages and difficulties of such implementations, and the literature gaps in this field.

3.4.2 Search Strategy

I developed a thorough search strategy using keywords such as ***"green roofs," "BIM," "sustainability," "modern architecture," "architecture design," and "case study."*** This strategy ensured that I captured a wide range of relevant studies from multiple perspectives.

	Keywords	Number of Papers
1	Green Roofs AND BIM	428
2	Green Roofs AND Modern Architecture	150
3	Green roofs AND Sustainability	3163
4	Green Roofs AND Architectural Design	718
5	Building Information Modeling AND Sustainability	3457
7	Building Information Modeling AND Modern Architecture	62
8	Green Roofs AND Building Information Modeling AND Case studies	139
9	Green Roofs AND BIM	428
10	Green Roofs AND Modern Architecture	150
	Grand Total	**8117**
	Screening of duplicates	*573*
	Deletion based on keywords	*3508*
	Included studies	**134**

Figure 15. Results from SLR (Author's Own,2024)

3.4.3 Database Searches

Searches were conducted across several major academic databases, *including Science Direct, Scopus, and Engineering Village*. These databases were chosen for their extensive collections of peer-reviewed articles and conference papers in the fields of engineering, architecture, and environmental science.

3.4.4 Screening and Selection Process

1. **Initial Search**: The initial search yielded approximately 8117 papers.
2. **Duplicate Removal**: After removing duplicates, the list was refined to 7400 papers.
3. **Title and Abstract Screening**: Titles and abstracts were screened to assess their relevance to the dissertation topic.
4. **Inclusion and Exclusion Criteria**: Papers were further filtered based on predefined inclusion and exclusion criteria. This rigorous process ensured that only the most pertinent studies were included.
5. **Final Selection**: The final list comprised 134 papers that were most relevant to my dissertation topic.

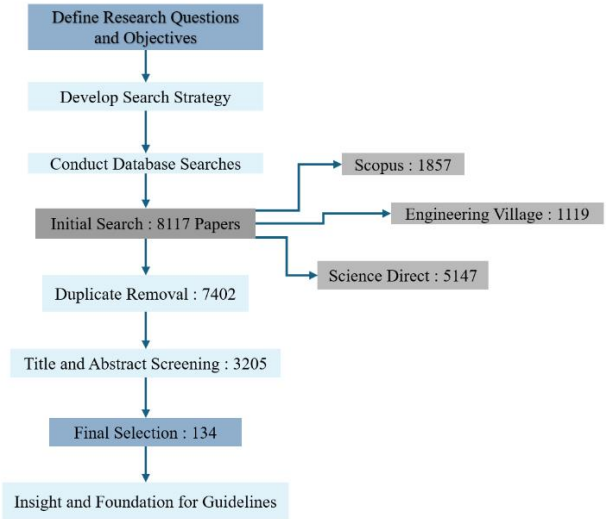


Figure 16.Initial Framework (Author's Own, 2024)

By adopting this systematic approach, I was able to ensure the comprehensiveness and reliability of my literature review. This method not only highlighted The present level of research on BIM and green roofs but also identified significant gaps and areas for future study. The insights gained from the SLR provided a solid foundation for the subsequent analysis and development of guidelines for integrating BIM in green roof projects(Fellows, 2015).

3.5 Ethical Considerations

For this dissertation, the main approach is the SLR, and the analysis of current case studies concerning the implementation of BIM in green roof design in contemporary architecture. This research does not propose any new primary data collection from human participants since it is a secondary data analysis of existing literature; thus, there are no ethical issues that can be associated with this study.

In this study, no interviews, surveys, or experiments that involve human subjects are conducted, therefore there is no need to discuss matters concerning informed consent, confidentiality, or harm to the participants. All the data used in this dissertation are collected from the publicly accessible databases which include Science Direct, Scopus, and Engineering Village hence the information used in this work is in the public domain and used in accordance with the ethical guidelines provided by these sources.

Also, issue of citation and referencing of all the used sources is well observed to enhance on the aspect of academic ethical consideration on plagiarism. In case of using case studies and previous research, the sources are always cited, and the intellectual property of other authors is observed and appreciated.

Thus, although ethical issues are a core component of research with human participants, they do not pertain to the research method used in this dissertation. This work complies with all the

rules of using secondary data and published sources, thus providing a solid and appropriate method of data analysis and synthesis.

Chapter 4: Benefits of BIM Integration in Green Roofs

4.1.1 Environmental Benefits

Sofronievska has pointed out that green roofs are useful in increasing energy efficiency in buildings. They perform an insulation function which saves on the use of air conditioning during the hot season and heating during the cold season. This natural insulation ability can cut down the energy usage, as a result, greenhouse gas emissions are reduced, while simultaneously, results in lower energy bills for the building owners (Carvalho et al., 2024).Additionally, the thermal mass by the soil and vegetation of green roofs also assist in regulating the internal temperatures; thus, making the buildings more efficient in energy usage(Sofronievska et al., 2024).

Besides, vegetation has the ability to reflect More light and absorbs less heat than the conventional roofing materials thus decreasing the urban heat island effect. (Carvalho et al., 2019).This not only helps in saving energy but also provides a better thermal environment to the (Carvalho et al., 2019).

4.1.2 Stormwater Management

Green roofs could serve a very important part in stormwater management especially in the urban areas. It is possible to store and hold large amounts of rainfall that in turn would decrease the flow of storm water into the drainage systems. Thus, it reduces the chances of over flooding and contributes to the decrease of the load on urban drainage systems (Rowe, 2011). Using BIM for green roof planning and development will help in proper management of stormwater by ensuring that appropriate kinds of vegetation and substrate are used to increase the roof's water retention and absorption capacities(Silva et al., 2023).

4.1.3 Urban Heat Island Mitigation

The UHI is a well-known phenomenon that poses an environmental hazard in most cities; The temperature in built-up zones is usually greater than that of the surrounding countryside due to anthropogenic heat discharge. This effect though is somewhat reversed by green roofs through cooling through evapotranspiration and shade.(Carvalho et al., 2019). The plants that are grown on green roofs help to store heat and release cool air to the surrounding environment hence reducing the temperature in urban areas. By integrating BIM, green roofs that are created can be modelled with great accuracy for the purpose of enhancing their ability to help in decreasing UHI, thus making urban areas cooler and more comfortable(Sofronievska et al., 2024).

4.2 Economic Benefits

4.2.1 Cost Savings

The economic benefits of green roofs include the ability to save energy costs because they provide insulation(Lee et al., 2024). Due to their insulating property, green roofs help in minimizing the use of heating and cooling systems, thus, leading to a reduction in energy costs for the building owner. BIM makes it easier for the estimated of these savings through Modelling of energy efficiency and determination of the best practices of green roof(Feng and Hewage, 2018).

4.2.2 Long-Term Maintenance Savings

Some of the disadvantages of green roofs include high initial costs than the conventional roofs but the costs of maintaining green roofs is relatively low. Thus, green roofs are relatively durable structures because they protect the other materials of the roof from various weather conditions and direct sunlight exposure(Nektarios, 2018). Because of these advantages, BIM is beneficial in managing and planning the maintenance of green roofs. Through the right and accurate Modelling of green roofs, BIM ensures the correct care of the roofs thus increasing the long-term profitability(Silva et al., 2015). Through the scheduling of the maintenance activities and real time tracking of the green roofs, BIM can be thus seen as very effective in increasing the life span of the green roofs and lowering the overall cost of ownership and thus being a useful tool in the formulation of long term financial plans(Sofronievska et al., 2024).

4.3 Social Benefits

4.3.1 Improved Aesthetics

Green roofs significantly enhance the appearance of buildings, urban environments, turning the conventional grey roofs into beautiful greenery settings. This change enhances the general beauty of the cities hence making them appealing and welcoming(Loder, 2014). The enhancement of green roofs is best facilitated using BIM since it helps architects and planners to develop an accurate description of green roofs. These visualisations properly convey the aesthetic value to the stakeholders and the public, on how green roofs can enhance the overall look of cities and other urban structures as it is more sustainable(Koscikova and Krivtsov, 2023).

4.3.2 Enhanced Urban Biodiversity

Green roofs offer suitable homes for plants and animals that are crucial in enhancing the ecological diversity in towns. These green spaces can support various forms of plant and animal life such as insects, birds, and small mammals thus creating small ecosystems on rooftops(Benvenuti, 2014). BIM can be used in designing green roofs with specific objectives of supporting species and biodiversity, thereby enhancing the ecological value. With the help of BIM, architects and planners can create accurate designs that enhance the habitats for the desired species and create diverse urban environments(Köhler and Ksiazek-Mikenas, 2018).

Chapter 5: Case Study Analysis

In the comparative case study analysis of The Edge in the Netherlands and Lagg Distillery in Scotland, several aspects will be explored. These include the selection criteria focusing on environmental impact and innovative design. The context and background of each project will be detailed, providing insight into their development and objectives. The analysis will cover BIM tools and technologies used, highlighting the software platforms and integration methods. Furthermore, the evaluation of green roof performance will consider design accuracy and project efficiency. Integration challenges will also be discussed, identifying both technical and operational challenges, along with solutions and best practices implemented. This comparative study seeks to highlight research gaps and develop a methodology for efficiently incorporating BIM into green roof projects.

5.1 Case 1: Lagg Distillery

5.1.1 Context and Background

The Lagg Distillery, located on the Isle of Arran, opened in 2019 and is designed to blend seamlessly with its natural surroundings. The project aimed to create an iconic landmark that reflects the island's heritage while promoting sustainability.(Newsdesk, 10 Feb, 2022)

5.1.2 Selection Criteria

Figure 17.Lagg Distillery

The reason why the case study of The Edge in Amsterdam was selected includes the project's environmental effect and its originality in design. The objective of the project was to minimize the adverse effects on the environment through the implementation of measures and concepts that are unique and eco-friendly. Finally, the integration of the green roofs was essential since they also played a dual role of providing aesthetics and green technology such as energy efficiency, biodiversity, and storm water management. These two concepts of sustainability and in-novation that have been described above were well incorporated in the design of the Edge, and for this reason it can be said that it is the perfect example of the modern sustainable architecture.

5.1.3 BIM Tools and Technologies Used

- **Software**: The architectural, engineering and construction sections were done in AutoCAD, whereas Revit was used to design the structure and the clash detection, and construction sequencing was done in Navisworks.
- **Integration Methods**: Used BIM in the design of the roof structure and geometry of the building, condition Modelling and regarding the environment and plant location.

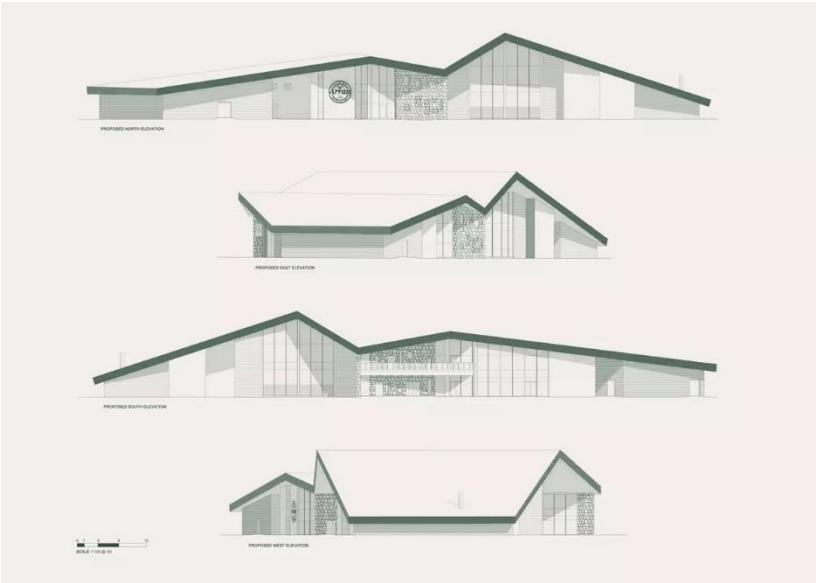


Figure 18.Profile of Lagg Distillery

5.1.4 Evaluation of Green Roof Performance

- **Design Accuracy**: High precision in modelling complex gradients and selecting plants resilient to extreme weather.
- **Project Efficiency**: Efficient management through precise coordination and real-time updates during construction, despite logistical challenges (Denham Youd, n.d.)
- .

5.1.5 Integration Challenges and Solutions

- **Technical Challenges**: Complex roof shapes, extreme weather conditions, logistical issues with material transport.
    - **Solutions**: Detailed BIM coordination and adaptive safety measures.
- **Operational Challenges**: Ensuring long-term maintenance in harsh weather, implementing safety measures during installation.
    - **Solutions**: BIM-driven maintenance scheduling and real-time performance monitoring. (Denham Youd, n.d.)

5.2 Case 2: The Edge, Netherlands

5.2.1 Selection Criteria

The Edge in Amsterdam was selected because of its acute application of innovation and efficiency in design for sustainability, having acquired the highest BREEAM rating and being ranked potentially as the world's smartest office building by Bloomberg(Carl, 2023).

Figure 19. Interior of The Edge, Netherlands

5.2.2 Context and Background

The Edge which is owned by OVG Real Estate and designed by PLP Architecture for Deloitte was launched in May 2015. This, ninety thousand square feet building incorporates state of the art technologies with the concept of smart building to enhance sustainability, technology, workforce, flexibility, and user satisfaction. The project was started in the year 2007 and then got stalled because of the economic downturn but was again taken up in the year 2010 with the view of coming up with a sustainable techno savvy office building(Carl, 2023).

5.2.3 BIM Tools and Technologies Used

Software and Platforms

- **Autodesk Revit**: Used for detailed architectural and structural modelling.
- **Navisworks**: Utilized for clash detection and construction sequencing.
- **Philips Connected Lighting System**: Integrated smart lighting system powered by Ethernet, allowing for individualized control and real-time data feedback(Carl, 2023).

Integration Methods

- **Detailed Modelling**: BIM was used to accurately model the complex structures, including the north-facing atrium and various green roof components.
- **Environmental Data Integration**: Simulated environmental conditions such as daylight optimization, energy efficiency, and aquifer thermal energy storage to ensure sustainable performance.
- **Smart System Integration**: Implemented An extensive system of sensors and advanced technologies that monitor and control lighting, climate, and other building systems(Carl, 2023).

5.2.4 Evaluation of Green Roof Performance

Design Accuracy

- **High Precision**: Achieved through detailed modelling of complex structures and integration of sustainable technologies, ensuring optimal plant selection and maintenance planning.

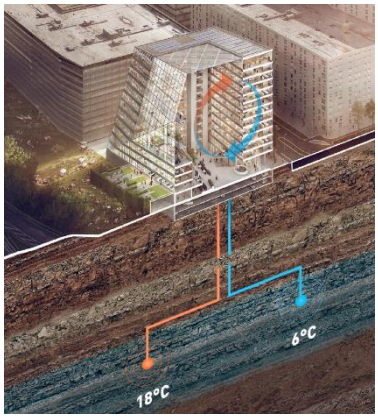
- **Daylight Optimization**: The design maximized daylight penetration while minimizing solar gain, improving workplace performance and reducing reliance on artificial lighting.

Project Efficiency

- **Efficient Management**: BIM facilitated efficient project management, reducing construction errors and ensuring timely completion despite economic challenges.
- **Energy Efficiency**: The building uses 70% less energy compared to similar buildings due to efficient massing, lighting, and climate control systems, contributing to its carbon-neutral status.(Carl, 2023)

5.2.5 Integration Challenges and Solutions

Technical Challenges



- **Smart System Integration**: Ensuring effective communication and interoperability among various smart systems posed significant technical challenges.
- **Complex Structures**: Modelling and constructing the complex roof shapes and integrating green roofs with other building systems required advanced BIM techniques.

Operational Challenges

- **User Adoption**: Encouraging users to fully utilize the smart systems and applications was challenging, as some preferred traditional methods.
- **Data Privacy**: Maintaining privacy while collecting extensive data from the building's sensors required robust data management practices.

Solutions and Best Practices

- **Close Collaboration**: Worked closely with technology suppliers to push the boundaries of existing solutions, resulting in 21 innovative implementations.
- **User-Centric Design**: Focused on designing flexible workspaces and user-friendly interfaces to enhance adoption and satisfaction.
- **Continuous Feedback**: Established a feedback loop to continuously monitor and improve system performance, ensuring long-term sustainability and user engagement.(Carl, 2023)

5.3 Case 3: Forester hill Health Campus, Aberdeen

5.3.1 Selection Criteria

The Forester hill Health Campus in Aberdeen was chosen for its significant focus on integrating green roofs into healthcare facilities. The project aimed to enhance patient recovery and well-being through sustainable design, providing a therapeutic environment that blends with the natural landscape.

5.3.2 Context and Background

The Forester Hill Health Campus is one of the largest healthcare complexes in Europe, which applies the green roofs to enhance people's health. It was intended that the green roofs would facilitate a more calming environment for the patients, and other members of the hospital, thus positively influencing the patients' health.(Scotland, Dec 1, 2016)

Figure 20.Forest Hill Campus

5.3.3 BIM Tools and Technologies Used

Software and Platforms

Autodesk Revit: Utilized for detailed architectural and structural modelling, allowing precise design and integration of green roofs.

Navisworks: Employed for clash detection and construction sequencing, ensuring the effective integration of green roof elements with existing infrastructure.

Integration Methods

Detailed Modelling: BIM facilitated the precise modelling of green roof layers, ensuring accurate design and integration.

Environmental Simulations: Conducted to optimize plant selection and ensure the long-term sustainability of the green roofs.

Construction Sequencing: Navisworks enabled the coordination of construction activities, minimizing disruptions and ensuring timely completion.(Scotland, Dec 1, 2016)

5.3.4 Evaluation of Green Roof Performance

Design Accuracy

High Precision: BIM allowed for accurate modelling of the green roofs, supporting diverse plant species suitable for the healthcare environment.

Sustainable Design: The green roofs were meant to be both sustainable and minimal maintenance, contributing to the long-term success of the project.

Project Efficiency

Efficient Coordination: BIM facilitated precise coordination among stakeholders, reducing errors and delays.

Timely Completion: Real-time updates and monitoring ensured that the project stayed on schedule and within budget.

5.4.5 Integration Challenges and Solutions

Technical Challenges

Long-term Maintenance: Ensuring the sustainability of the green roofs and integrating them with existing healthcare infrastructure.

Solutions: Implemented BIM-driven maintenance scheduling and real-time performance monitoring to address issues promptly.

Operational Challenges

Healthcare Environment: Maintaining green roofs without disrupting hospital operations and ensuring patient access.

Solutions: Careful planning and coordination using BIM tools ensured that green roofs were maintained without affecting hospital activities. Adaptive safety measures were also implemented during installation.

Solutions and Best Practices

BIM-Driven Maintenance: Regular maintenance scheduling and performance monitoring through BIM ensured the green roofs remained sustainable and functional.

Adaptive Design: Designing green roofs that were accessible yet safe for patients and staff helped integrate these spaces seamlessly into the healthcare environment.

5.4 Case 4: Bloomberg, London

5.4.1 Context and Background

The Bloomberg London headquarters by Foster + Partners located in the City of London is one of the most excellent sustainable office designs. It was intended that the construction of the building would establish new standards in matters of sustainability, efficiency and comfort of the occupants. It applies a well evolved system of technologies and design strategies to minimize the building's environmental effect. The selection criteria for the project priorities were established to achieve the highest environmental standards, and this is evidenced by the BREEAM Outstanding rating that marks the highest score for a major office project at the design stage (AECOM, 2017),

5.4.2 BIM Tools and Technologies Used

Software and Platforms

The Bloomberg London headquarters utilized a range of BIM tools and technologies to achieve its sustainability and efficiency goals. Key software platforms included Autodesk Revit for architectural and structural Modelling, Navisworks for project review and clash detection, and Dynamo for computational design and automation. These tools allowed the design team to create detailed 3D models, perform simulations, and optimize the building's performance in various environmental conditions(AECOM, 2017).

Integration Methods: BIM was used throughout the project from conception to design, construction and through to the facilities management. It helped to improve communication between architects, engineers, contractors, and facility managers and their interconnection. BIM application enabled the revision and updating of the building model, coordinating all the changes to the design and meeting the project's high sustainability standards. Also, the digital twin developed through the use of BIM allowed for real-time control of the facility's systems when the building was complete (AECOM, 2017).

5.4.3 Evaluation of Green Roof Performance

Design Accuracy The green roof at Bloomberg London was designed to provide significant environmental benefits, including improved insulation, stormwater management, and biodiversity enhancement. BIM tools were crucial in achieving a high level of design accuracy, ensuring that the green roof system was optimally integrated with the building's structure and other environmental systems. This precision helped to maximize the performance and effectiveness of the green roof, contributing to the overall sustainability of the building (Partners, 2018).

Project Efficiency BIM enhanced the aspects of project delivery by providing a solution to design and construction procedures. This helped in planning and coordination that advocated the use of BIM to avoid errors and repetition of work. This was especially critical for the installation of the green roof that demanded precise connection to the building's structure and water management systems. The project was delivered on schedule and to the cost, which

proves the efficiency of BIM in organizing large and environmentally friendly construction projects (Partners, 2018)

5.4.4 Integration Challenges and Solutions

Technical Challenges One of the primary technical challenges was ensuring the seamless integration of the green roof with the building's other advanced systems, including its natural ventilation and water management systems. The complexity of the design required precise Modelling and simulation to ensure that all components worked together effectively. Additionally, the building's location in a dense urban environment posed challenges related to structural loading and maintenance access (AECOM, 2017).

Operational Challenges Operational challenges included maintaining the green roof's effectiveness over time ensuring that it continued to deliver environmental benefits. Regular maintenance and monitoring were necessary to manage plant health, irrigation needs, and drainage performance. Ensuring that the building's occupants were engaged and aware of the green roof's benefits also posed a challenge (AECOM, 2017).

Solutions and Best Practices To overcome these challenges; the project team adopted the following recommendations. Some of the measures that were taken were the application of high-end BIM for the design and Modelling of the structure, constant interaction between all the concerned entities, and the formulation of a strict maintenance strategy. The project also incorporated real time data logging for the green roof and other systems and using this data the project team was able to monitor and maintain the systems effectively. These measures made sure that the green roof and the other sustainable strategies achieved their desired objectives and raised the bar for sustainable office design (Partners, 2018).

5.5 Comparative Analysis

Criteria	Bloomberg, London	The Edge, Amsterdam	Lagg Distillery, Isle of Arran	Forester hill Health Campus, Aberdeen
Year Built	2017	2015	2019	2018
Architect Designed	Foster + Partners	PLP Architecture	Organic Architects	Mackie Ramsay Taylor Architects
Built By	Sir Robert McAlpine	OVG Real Estate	Robertson Construction	Morrison Construction
BIM Tools and Technologies Used	Autodesk Revit, Navisworks	BIM 360, Revit	Industry-standard tools	Revit, Navisworks
Integration Methods	Energy simulations, structural analysis, integration with sustainable features (Global Construction Review, 2017)	IoT integration for real-time data management, environmental control (Eco-Business, 2016)	Project coordination, sustainability planning	Design, construction, facility management (Panorama Solutions, 2021)
Performance Metrics	BREEAM 98.5%, 35% energy savings (Global Construction Review, 2017)	BREEAM 98.36%, high energy savings (PLP Architecture, 2015)	Local sustainability targets	High healthcare sustainability goals (Thomson Gray, 2018)
Level of Development (LOD)	LOD 500	LOD 400-500	LOD 300-400	LOD 500
Materials Used	LED lights, bronze blades, recycled water (DesignCurial, 2017)	High-performance glazing, automated blinds, advanced HVAC (Eco-Business, 2016; ArchDaily, 2015)	Locally sourced sustainable materials	Specialized healthcare materials, green roofs (Thomson Gray, 2018)
Time Taken	~5 years	~3 years	~2 years	~4 years
Energy Efficiency	Significant reduction in energy consumption (35%) (Global Construction Review, 2017)	Optimized through advanced energy management systems (PLP Architecture, 2015)	Achieved through sustainable practices	Enhanced through integrated systems (Panorama Solutions, 2021)
Stormwater Management	Advanced rainwater harvesting systems	Effective through green roof integration	Localized water management	Green roofs enhance water management
Urban Heat Island Mitigation	Effective through green roof and materials used	High impact due to extensive green roofs and reflective materials	Moderate impact through design	Significant due to healthcare facility needs
Design Accuracy	High due to BIM integration (Global Construction Review, 2017)	Precise through BIM and IoT (Eco-Business, 2016)	Moderate, focusing on local sustainability	High, customized for healthcare (Thomson Gray, 2018)
Project Efficiency	Increased due to BIM coordination	High due to advanced BIM and IoT (PLP Architecture, 2015)	Improved through BIM planning	Enhanced by BIM for healthcare (Panorama Solutions, 2021)
Technical Challenges	Integration with existing infrastructure (ArchDaily, 2017)	Complex BIM-IoT integration (Eco-Business, 2016)	Ensuring local practices in BIM	Healthcare-specific requirements (Thomson Gray, 2018)
Operational Challenges	Long-term sustainability maintenance	Maintaining optimal performance	Operational sustainability	High standards of healthcare operations
Solutions and Best Practices	Modular design, continuous BIM monitoring	Real-time data management, adaptive systems	Local materials, sustainable practices	Customized BIM solutions for healthcare

5.5.1 Results of Comparative Analysis

The study brings out how BIM has changed the design, construction, as well as the management of green roof projects. With the focus on certain environmental, economic and social objectives, these projects demonstrate how versatile and beneficial BIM is in any context. Technologies applied, materials chosen, and the management of the project thanks to the BIM helped to achieve many sustainable gains.

The research outcomes highlight the need of using BIM in green roof projects as a way of enhancing the performance, efficiency and sustainability. This study can inform future projects as to how they can avoid some of the pitfalls and get the most out of BIM for enhancing urban resilience and sustainability. This complete model provides a general guideline for architects, engineers and construction managers on how to incorporate BIM in green roof projects to bring about the much needed environmental and economic change.

Chapter 6: Development of a BIM Integration Framework

BIM has a strong potential to optimize environmental, economic, and social aspects of green roof projects when applied to the design of such construction. This chapter provides a comprehensive plan to improve the design, installation, and maintenance of green roofs through the use of BIM technology. The framework is developed to help architects, engineers, construction managers, and other parties concerned in green roof projects. Thus, the use of BIM in the framework helps to implement green roof installations that positively affect the urban resilience and the sustainability targets. The procedures and the elements that are needed for the integration are outlined in the following guidelines.

6.1. Goals and Scope

The overall goal of this framework is to present strategies for enhancing the application of BIM in green roofs design and construction in the built environment. In an effort to enhance the integration of advanced BIM tools in the provision of better environmental performance, economic efficiency and social wellbeing of the framework. This book is suitable for architects, engineers, construction managers, and other participants in green roofs projects. Within the context of this paper, strategies pertain to green roof projects that have BIM for designing and constructing as well as assessing the improvements that the project has on the urban resilience and sustainability.

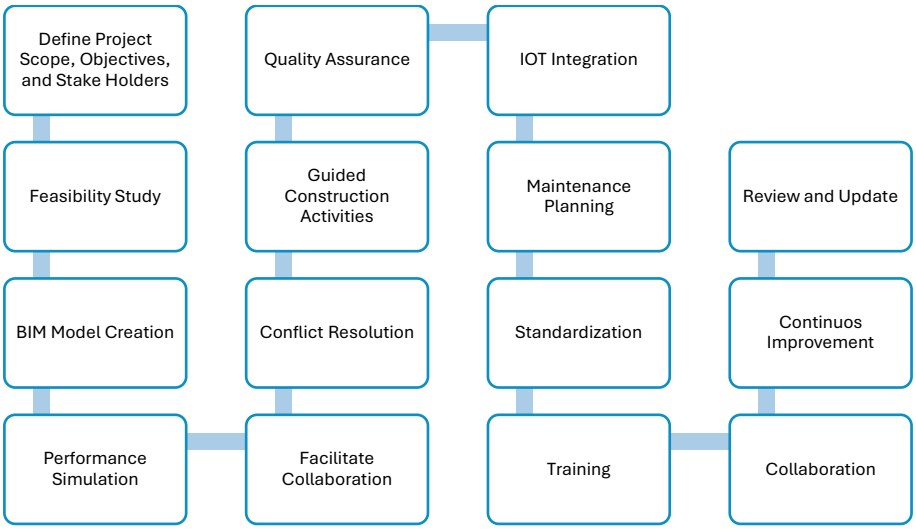


Figure 21.. BIM Integration Framework (Author's Own,2024)

6.2 Framework Guidelines

1. Define Project Scope, Objectives, and Stake Holders: First, it is crucial to set the objectives and the tasks that the project will accomplish, the project's area of interest, and the expected outcome of the project in addition to defining the people who will be impacted by the project.

It is important to clarify the expectations and the duties of each member of the team to prevent misunderstandings and discrepancies within the team.

2. Feasibility Study: It is recommended that a detailed analysis of the feasibility of the proposed plan in terms of technical and economic efficiency is carried out. Assess the environmental factors and resources that are required to define the possible risks that may affect the success of the project.

3. BIM Model Creation: Prepare a comprehensive schedule of the BIM models which have architectural, structural and mechanical, electrical and plumbing (MEP) objects. Software must be up to date and the model must be elaborated with flexibility to prevent problem in the later stages of the project.

4. Performance Simulation: They can use the BIM model to try and model different scenarios of events and analyse the probable performance and then come up with a way of how to work on it. These simulations are useful in informing decisions that can be made on the design aspect of a building to ensure it is efficient when it comes to energy use, structural stability and the effects on the environment.

5. Facilitate Collaboration: Integrate tools that will enable the real time communication and sharing of information with the various stakeholders through the BIM software. This makes sure that every member is informed of the expectations and thus aids in avoiding cases of different goals and objectives of the project as well as delays.

6. Conflict Resolution: Applies the clash detection features of the BIM to detect and solve design interferences before the construction phase to prevent expensive and time-wasting alterations during construction.

7. Guided Construction Activities: The BIM model should be used in construction to check whether the construction activities being carried out are in accordance with the planned specifications and designs.

8. Quality Assurance: Rarely check the construct work against BIM model for the quality assurance during the construction activity. This measure is taken to avoid poor construction work which may need costly corrections in the future.

9. IoT Integration: Improve the management of the project and control through the use of IoT devices that would feed real time information which is very helpful in decision making and management of the project.

10. Maintenance Planning: Apply the data and findings from BIM and IoT for proper scheduling and maintenance of the building to meet future needs and improve the building's performance.

11. Standardization: Adopt and adhere to BIM standard approaches for the entire project and all other related projects to achieve best practices. Don't use custom standards as they are likely to cause issues in collaborating and integrating the processes.

12. Training: It is recommended to train all the project team members in the usage of BIM technologies to increase the efficiency of the utilizing technologies. Continued education reduces on the occurrence of errors and improves on the project quality.

13. Collaboration: It is necessary to create the condition under which all the participants will be able to provide their input and suggestions on a regular basis. This is a very important aspect since it helps in solving problem that may occur from time to time and also encourage innovation.

14. Continuous Improvement: Continually reassess and optimise processes and technologies according to project's data and new tendencies in the industry to make the project more competitive.

15. Review and Update: Revise and elevate the BIM model and project strategies from time to time with the most recent trends and remarks to make them useful and efficient.

6.3 Framework Components

6.3.1 Key Elements and Tools

BIM Software: According to the framework, it is suggested to utilize the conventional BIM software such as *Revit from Autodesk and Navisworks*. These tools are very useful for developing the detailed architectural designs, energy simulations, and structural analysis. BIM software is used to model the green roof components in a proper manner, and one can be able to mimic their behaviour in different climates as well as easily incorporate them into the building model. This is useful in determining the load bearing capability, drainage systems as well as the insulation properties of the structure. Through incorporation of BIM, the design aspect is made to be more precise, construction errors are reduced, and the collaboration of the various parties is well coordinated. The visualization tools of the BIM help the project teams visual the green roof in relation to the rest of the structure and helps in decision making and minimizes on changes that may be costly in the future.

Green Roof Design: Several factors like, type and species of vegetation to be planted, the depth of the soil layer to be used, type of drainage system to be incorporated and the irrigation system to be used among others have to be put into consideration in order to achieve the intended objectives of constructing a green roof. Thus, the framework comprises of these elements and how they ought to be developed to be incorporated into the BIM models and to be efficient. BIM is implemented to develop a scale model of the green roofs' system's layers such as the waterproofing membrane, root barrier, drainage layer, growing medium, plants, and the effect of green roof on the structural system of the building. Thus, green roof design in BIM models assists in formulation of the design and performance of the roof. These aspects result in the storm water system being upgraded, the UHI effect is minimized, and energy performance of

the building is increased. The features can be realized due to the fact that BIM allows for the detailed analysis and the modeling of the performance at the design stage.***Data Integration:*** The framework also promotes the use of Internet of Things (IoT) sensors to gather current data on the environment including moisture, temperature and health of the plants. These sensors are linked with BIM system that enable the process of monitoring and gathering of feedback data. To update the simulation models, input data from IoT sensors is incorporated into the BIM model. This integration assists in modifying the schedules of maintenance, determining the amounts of irrigation required and the condition of the green roof. The application of IoT with BIM greatly improves the inspection and management of green roof infrastructures. It offers preventive measures, minimises the possibilities of system failures, improves on resource management, and offers a wealth of data that can be useful in future undertakings through documentation.

6.4 Expected Outcomes and Benefits

6.4.1 Environmental Benefits

Energy Efficiency: Green roofs can also enhance insulation of the buildings thus utilizing less energy in heating and cooling. BIM simulations offer the following benefits and Modelling different design scenarios can enhance these benefits.

Stormwater Management: Optimal design and management of green roofs through the use of BIM can help in reducing stormwater runoff and therefore decrease the incidences of floods and enhance water quality.

Urban Heat Island Mitigation: It also reduces the temperature in urban areas which can be designed and optimized with high precision using BIM.

6. 4. 2 Economic Benefits

Cost Savings: The use of BIM improves the design and construction process and minimize the occurrences of errors and its consequences there from. Thus, the long-term profits are lower energy consumption and maintenance expenditures.

Increased Property Value: Green roofs are built on properties and improve the value of the property; this is because of the aesthetic value and the enhanced sustainability feature.

6.4.3 Social Benefits

Improved Health and Well-being: They offer recreational areas and enhance the quality of air thus enhancing the health of the occupants of the building as well as the surrounding society.

Educational and Community Engagement: Such roofs can be useful in educating people in sustainable practices and can also bring the community together through gardening and recreational space.

Chapter 7: Discussion and Conclusion

This chapter provides a summary of the main conclusions of the research and connects the outcomes from the systematic literature review, case studies and the implementation of BIM in green roof projects. This discussion will assess the significance of BIM in improving the design, construction and sustainability of green roofs to the research questions stated at the beginning of this study. Also, this chapter will discuss the consequences of these findings for the overall subject of sustainable urban architecture and define the advantages and drawbacks of applying BIM. The final chapter will present the overall findings of the research and suggest directions for further research and practice on the enhancement of BIM for green roofs, as well as the future developments like smart technologies and advanced simulations. Thus, within the framework of this extensive analysis, the chapter intends to provide practitioners and researchers with a solid tool to improve the performance of urban environments in a more sustainable manner.

7.1 Analysis of Findings

7.1.1 Summary of Case Study Results

There are several projects that have successfully implemented BIM with green roofs and few of them are Bloomberg London, The Edge in Amsterdam, Lagg Distillery, and Forester hill Health Campus. All four cases revealed considerable advantages regarding sustainability, production, and the level of design.

i. Bloomberg London: It was able to attain a BREEAM Outstanding rating of 98. 5 percent with considerable cut down on energy use (35 percent). Autocad Revit and Navisworks helped in the detailed design, energy analysis, and structural analysis of the project which in turn played a positive role in the overall success of the project(AECOM, 2017).
ii. The Edge, Amsterdam: It is the world's greenest office building with BREEAM rating of 98%. 36%. Making use of BIM 360 and IoT in real time data management and environmental control of the building enhanced operational effectiveness and sustainability (Carl, 2023).
iii. Lagg Distillery: Being oriented on the local context, the planning and management of the project has been carried out with the help of standard BIM applications. The project also aimed at the incorporation of local materials and construction with consideration of environmentally friendly methods(Newsdesk, 10 Feb, 2022).
iv. Forester hill Health Campus: Realized high scores in the healthcare-industry related sustainability objectives with the aid of Revit and Navisworks for facility management and sustenance. The enhancement of green roofs enhanced the health care setting for the patients and the personnel (Scotland, Dec 1, 2016).

7.1.2 Interpretation of Data

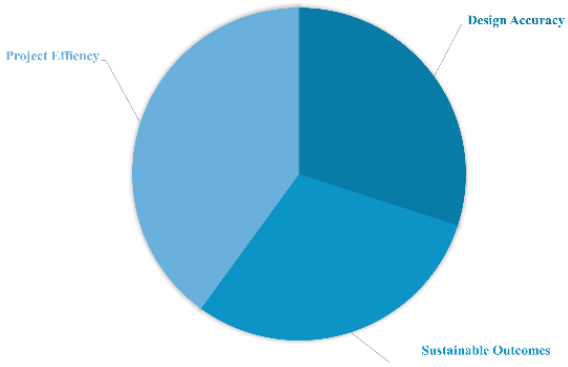


Figure 22.Impact of BIM on Green Roofs(Author's Own, 2024)

This study implies that BIM application in green roof projects improves design information reliability and project organization, as well as its sustainability. BIM enhances the possibility of planning and Modelling of the green roof and its performance. This is because the integration of IoT with BIM enhances real-time monitoring as well as the maintenance leading to the sustainability of green roof systems.

Altogether, these case studies demonstrate how BIM can be used to transform sustainable urban architecture through the identification of proper strategies for green roofs integration.

7.2 Implications for Architects and Urban Planners

7.2.1 Design Considerations

The following should be taken into consideration by architects and urban planners when designing the green roofs with BIM. The following design considerations will therefore be useful to the architects and urban planners when implementing green roofs with BIM.

1. ***Comprehensive Modelling:*** Get all the green roof details incorporating them into a BIM model so that vegetation, soil and drainage systems are incorporated. This for the purpose of highly accurate reproductions, and predictions in terms of performance.
2. ***Collaboration:*** Use BIM model to facilitate interaction between architects, engineers and contractors so that they can effectively collaborate. It is critical for this process to be collaborative since there may be major clashes in terms of design requirements that may only show up during planning if not undone early in the scheme's design phase.
3. ***Sustainability Goals:*** Design green roofs in a sustainable context, therefore connecting them with other general goals, including energy, stormwater and heat island reduction.

7.2.2 Policy Recommendations

In order to encourage the application of BIM-integrated green roofs, the following recommendations should be made to the policymakers. To encourage the implementation of BIM integrated green roofs the following recommendations should be made to the policy makers.

1. ***Incentives:*** Give credit to projects which adopt BIM and green roofs, including offering them the necessary tax exemptions or grants.
2. ***Standards and Regulations:*** Implement procedures for the use of BIM Modelling and its adoption of green roofs for future projects as well as the emergence of standard guidelines that will be followed to maintain order when dealing with both elements.

3. ***Training and Education:*** Promote awareness of that to undertake training programs for architects, engineers, and all other construction specialists on BIM and sustainable construction practices.

7.3 The Role of BIM in Future Green Roof Projects

7.3.1 Future Trends

Thus, it is seen that BIM is all slated to becoming even more indispensable in the future of green roof business. Several features that will boost the effectiveness and possibilities of BEM are arising, which will continue the advancement of green roofs in cities.

Advanced Simulations

Based on the future BIM advancement, there would be enhancements in the specific simulation of green roof functionality according to different climatic conditions. These advanced simulations shall enable the architect and engineer to determine how a green roof will perform under varying climatic conditions, changes in climate and city pollution. Advanced simulation tools will improve the specifics of green roof designs for energy, stormwater, and thermal performance resulting in better green roof systems.

Integration with Smart Technologies

The use of BIM in conjunction with the smart technology like IoT and AI will completely transform the management and maintenance of green roofs. IoT devices can give the real-time data about the moisture content, temperature, and health of the plants which can further be integrated with the BIM models for their monitoring. Thus, by employing usage of AI to forecast when and what part of the green roof will require maintenance, resources can be efficiently utilised to restore and maintain functionality of the green roofs. This integration of technologies shall lead to realization of preventive maintenance approach towards such systems and thus the increased the durability of green roofs.

Sustainability Metrics

The added sustainability features embedded in BIM will enhance multi-parametric measures of sustainable green roofs' environmental, economic and sociological effects. These will allow project teams to assess the results of the green roofs in relation to sustainable objectives such as emissions' cut, species' diversification, urban greenery expansion, etc. Some of these metrics include. By integrating such metrics into BIM models, the stakeholders would be able to take right decisions concerning; the green roofs design and management and thus improving their RVI. Also, these metrics can be used in obtaining the certification and under the required legislation policies that support the green roof system in the current urban development processes.

7.3.2 Potential Innovations

Future innovations in BIM and green roofs may include:

Automated Design: Design applications or features within BIM that use AI to calculate and automatically create green roofs following data collected from the site's conditions and sustainability goals.

Materials Innovation: The update of the BIM models with new sustainable materials for the purpose of providing options for new green roof ideas.

Community Engagement: Integration of several technical developments directly into BIM that make it easier for the authorities and local people to participate in green roof projects.

Thus, this paper has demonstrated how these trends and innovations work together to support BIM in further improving the design, implementation, and management of green roofs, therefore advancing the sustainability and development of urban systems.

7.4 Limitations of the Study

Scope of Case Studies:

The research is based on a rather small number of investigated cases, which can hardly be considered as fully representative of green roof practices worldwide. The selected cases are mainly from significant projects and large-scale green roofs which may hide the specifics and advantages of smaller green roofs or less known projects.

Data Availability:

The data from the case studies was sources were not uniform and the quality of data also differed. In this regard, some of the projects provided a lot of and detailed information while others furnished very little information. This discrepancy could affect the extent of the analysis as well as the transference of the results.

Technological Constraints:

Thus, the dynamic development of BIM and green roof technologies indicates that some of the tools and methods applied in this research may become ineffective shortly. The research was based on the existing technology, which cannot always predict the possibilities of a future development.

Geographical Focus:

The case studies are mainly based on projects in certain geographic areas and especially in the UK and Europe. This emphasis could restrict the generalization of the results to other zones characterized by different climatic, economic, and legal conditions.

Standardization Issues:

This can be a disadvantage because there are no set guidelines for the use of BIM, and this makes it difficult to compare different projects. This limitation may impact the validity of the comparative analyses that were done in the study.

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